SOLUTION OF HOMEWORK 07

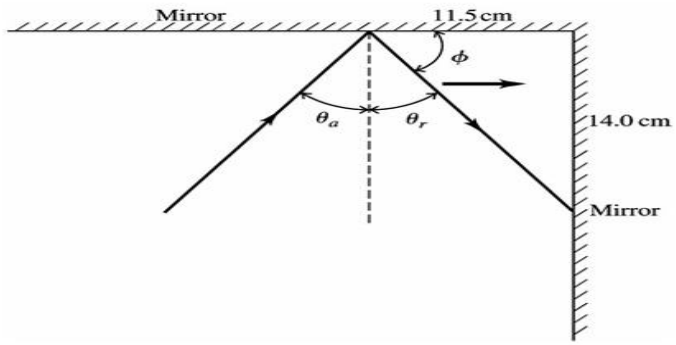
1.

IDENTIFY: For reflection, $\theta_r = \theta_a$.

SET UP: The desired path of the ray is sketched in Figure 33.1.

EXECUTE: $\tan\phi = \dfrac{14.0\text{ cm}}{11.5\text{ cm}}$, so $\phi = 50.6°$. $\theta_r = 90° - \phi = 39.4°$ and $\theta_r = \theta_a = 39.4°$.

EVALUATE: The angle of incidence is measured from the normal to the surface.



2.

IDENTIFY: The speed and the wavelength of the light will be affected by the vitreous humor, but not the frequency.

SET UP: $n = \dfrac{c}{v}$. $v = f\lambda$. $\lambda = \dfrac{\lambda_0}{n}$.

EXECUTE: **(a)** $\lambda_v = \dfrac{\lambda_{0,v}}{n} = \dfrac{400\text{ nm}}{1.34} = 299\text{ nm}$. $\lambda_r = \dfrac{\lambda_{0,r}}{n} = \dfrac{700\text{ nm}}{1.34} = 522\text{ nm}$. The range is 299 nm to 522 nm.

(b) Calculate the frequency in air, where $v = c = 3.00\times10^8$ m/s. $f_r = \dfrac{c}{\lambda_r} = \dfrac{3.00\times10^8\text{ m/s}}{700\times10^{-9}\text{ m}} = 4.29\times10^{14}$ Hz.

$f_v = \dfrac{c}{\lambda_v} = \dfrac{3.00\times10^8\text{ m/s}}{400\times10^{-9}\text{ m}} = 7.50\times10^{14}$ Hz. The range is 4.29×10^{14} Hz to 7.50×10^{14} Hz.

(c) $v = \dfrac{c}{n} = \dfrac{3.00\times10^8\text{ m/s}}{1.34} = 2.24\times10^8$ m/s.

EVALUATE: The frequency range in air is the same as in the vitreous humor.

3.

IDENTIFY: The time delay occurs because the beam going through the transparent material travels slower than the beam in air.

SET UP: $v = \dfrac{c}{n}$ in the material, but $v = c$ in air.

EXECUTE: The time for the beam traveling in air to reach the detector is

$t = \dfrac{d}{c} = \dfrac{2.50\text{ m}}{3.00\times10^8\text{ m/s}} = 8.33\times10^{-9}$ s. The light traveling in the block takes time

$t = 8.33\times10^{-9}\text{ s} + 6.25\times10^{-9}\text{ s} = 1.46\times10^{-8}$ s. The speed of light in the block is

$v = \dfrac{d}{t} = \dfrac{2.50\text{ m}}{1.46\times10^{-8}\text{ s}} = 1.71\times10^8$ m/s. The refractive index of the block is $n = \dfrac{c}{v} = \dfrac{3.00\times10^8\text{ m/s}}{1.71\times10^8\text{ m/s}} = 1.75$.

EVALUATE: $n > 1$, as it must be, and 1.75 is a reasonable index of refraction for a transparent material such as plastic.

4.

IDENTIFY: Since the refractive index of the glass is greater than that of air or water, total internal reflection will occur at the cube surface if the angle of incidence is greater than or equal to the critical angle.

SET UP: At the critical angle θ_{crit}, Snell's law gives $n_{\text{glass}} \sin\theta_{\text{crit}} = n_{\text{air}} \sin 90°$ and likewise for water.

EXECUTE: **(a)** At the critical angle θ_{crit}, $n_{\text{glass}} \sin\theta_{\text{crit}} = n_{\text{air}} \sin 90°$.

$1.53 \sin\theta_{\text{crit}} = (1.00)(1)$ and $\theta_{\text{crit}} = 40.8°$.

(b) Using the same procedure as in part (a), we have $1.53 \sin\theta_{\text{crit}} = 1.333 \sin 90°$ and $\theta_{\text{crit}} = 60.6°$.

EVALUATE: Since the refractive index of water is closer to the refractive index of glass than the refractive index of air is, the critical angle for glass-to-water is greater than for glass-to-air.

5.

IDENTIFY: Use the critical angle to find the index of refraction of the liquid.

SET UP: Total internal reflection requires that the light be incident on the material with the larger n, in this case the liquid. Apply $n_a \sin\theta_a = n_b \sin\theta_b$ with $a =$ liquid and $b =$ air, so $n_a = n_{\text{liq}}$ and $n_b = 1.0$.

EXECUTE: $\theta_a = \theta_{\text{crit}}$ when $\theta_b = 90°$, so $n_{\text{liq}} \sin\theta_{\text{crit}} = (1.0)\sin 90°$

$$n_{\text{liq}} = \frac{1}{\sin\theta_{\text{crit}}} = \frac{1}{\sin 42.5°} = 1.48.$$

(a) $n_a \sin\theta_a = n_b \sin\theta_b$ ($a =$ liquid, $b =$ air)

$$\sin\theta_b = \frac{n_a \sin\theta_a}{n_b} = \frac{(1.48)\sin 35.0°}{1.0} = 0.8489 \text{ and } \theta_b = 58.1°$$

(b) Now $n_a \sin\theta_a = n_b \sin\theta_b$ with $a =$ air, $b =$ liquid

$$\sin\theta_b = \frac{n_a \sin\theta_a}{n_b} = \frac{(1.0)\sin 35.0°}{1.48} = 0.3876 \text{ and } \theta_b = 22.8°$$

EVALUATE: For light traveling liquid $\rightarrow$ air the light is bent away from the normal. For light traveling air $\rightarrow$ liquid the light is bent toward the normal.

6.

IDENTIFY: Snell's law is $n_a \sin\theta_a = n_b \sin\theta_b$. $v = \dfrac{c}{n}$.

SET UP: $a =$ air, $b =$ glass.

EXECUTE: **(a)** red: $n_b = \dfrac{n_a \sin\theta_a}{\sin\theta_b} = \dfrac{(1.00)\sin 57.0°}{\sin 38.1°} = 1.36$. violet: $n_b = \dfrac{(1.00)\sin 57.0°}{\sin 36.7°} = 1.40$.

(b) red: $v = \dfrac{c}{n} = \dfrac{3.00\times10^8 \text{ m/s}}{1.36} = 2.21\times10^8 \text{ m/s}$; violet: $v = \dfrac{c}{n} = \dfrac{3.00\times10^8 \text{ m/s}}{1.40} = 2.14\times10^8 \text{ m/s}$.

EVALUATE: n is larger for the violet light and therefore this light is bent more toward the normal, and the violet light has a smaller speed in the glass than the red light.

7.

IDENTIFY: When unpolarized light passes through a polarizer the intensity is reduced by a factor of $\frac{1}{2}$ and the transmitted light is polarized along the axis of the polarizer. When polarized light of intensity $I_{\max}$ is incident on a polarizer, the transmitted intensity is $I = I_{\max}\cos^2\phi$, where ϕ is the angle between the polarization direction of the incident light and the axis of the filter.

SET UP: For the second polarizer $\phi = 60°$. For the third polarizer, $\phi = 90° - 60° = 30°$.

EXECUTE: **(a)** At point A the intensity is $I_0/2$ and the light is polarized along the vertical direction. At point B the intensity is $(I_0/2)(\cos 60°)^2 = 0.125 I_0$, and the light is polarized along the axis of the second polarizer. At point C the intensity is $(0.125 I_0)(\cos 30°)^2 = 0.0938 I_0$.

(b) Now for the last filter $\phi = 90°$ and $I = 0$.

EVALUATE: Adding the middle filter increases the transmitted intensity.

8.

IDENTIFY: The reflected light is completely polarized when the angle of incidence equals the polarizing angle θ_p, where $\tan\theta_p = \frac{n_b}{n_a}$.

SET UP: $n_b = 1.66$.

EXECUTE: **(a)** $n_a = 1.00$. $\tan\theta_p = \frac{1.66}{1.00}$ and $\theta_p = 58.9°$.

(b) $n_a = 1.333$. $\tan\theta_p = \frac{1.66}{1.333}$ and $\theta_p = 51.2°$.

EVALUATE: The polarizing angle depends on the refractive indicies of both materials at the interface.

9.

IDENTIFY: Use the transmitted intensity when all three polairzers are present to solve for the incident intensity I_0. Then repeat the calculation with only the first and third polarizers.

SET UP: For unpolarized light incident on a filter, $I = \frac{1}{2} I_0$ and the light is linearly polarized along the filter axis. For polarized light incident on a filter, $I = I_{\max}(\cos\phi)^2$, where $I_{\max}$ is the intensity of the incident light, and the emerging light is linearly polarized along the filter axis.

EXECUTE: With all three polarizers, if the incident intensity is I_0 the transmitted intensity is $I = (\frac{1}{2} I_0)(\cos 23.0°)^2(\cos[62.0° - 23.0°])^2 = 0.256 I_0$. $I_0 = \frac{I}{0.256} = \frac{75.0\ \text{W/cm}^2}{0.256} = 293\ \text{W/cm}^2$. With only the first and third polarizers, $I = (\frac{1}{2} I_0)(\cos 62.0°)^2 = 0.110 I_0 = (0.110)(293\ \text{W/cm}^2) = 32.2\ \text{W/cm}^2$.

EVALUATE: The transmitted intensity is greater when all three filters are present.

10.

IDENTIFY: The shorter the wavelength of light, the more it is scattered. The intensity is inversely proportional to the fourth power of the wavelength.

SET UP: The intensity of the scattered light is proportional to $1/\lambda^4$; we can write it as $I = (\text{constant})/\lambda^4$.

EXECUTE: **(a)** Since I is proportional to $1/\lambda^4$, we have $I = (\text{constant})/\lambda^4$. Taking the ratio of the intensity of the red light to that of the green light gives

$$\frac{I_R}{I} = \frac{(\text{constant})/\lambda_R^4}{(\text{constant})/\lambda_G^4} = \left(\frac{\lambda_G}{\lambda_R}\right)^4 = \left(\frac{520\ \text{nm}}{665\ \text{nm}}\right)^4 = 0.374, \text{ so } I_R = 0.374 I.$$

(b) Following the same procedure as in part (a) gives $\frac{I_V}{I} = \left(\frac{\lambda_G}{\lambda_V}\right)^4 = \left(\frac{520\ \text{nm}}{420\ \text{nm}}\right)^4 = 2.35$, so $I_V = 2.35 I$.

EVALUATE: In the scattered light, the intensity of the short-wavelength violet light is about 7 times as great as that of the red light, so this scattered light will have a blue-violet color.